

Old UNM Algebra Qualifier Problems

Based on Chapter 4 of Dummit–Foote

Fifteen-year collection: January 2012–August 2023

Scope. This packet uses the third edition of Dummit and Foote, *Abstract Algebra*. Chapter 4 is “Group Actions”. The first section contains problems that can reasonably be attempted after this chapter. The second section records closely related problems.

I. Core problems from Chapter 4

January 2012 Qualifier, Problem 1

How many elements of order 7 must be in a simple group of order 168?

January 2013 Qualifier, Problem 1

Let p be a prime. Show that every group with p^2 elements is abelian.

Note: We will see later that there are nonabelian group with p^3 elements.

January 2017 Qualifier, Problem 1

Suppose p and q are distinct primes such that

$$p \nmid q \pm 1 \quad \text{and} \quad q \nmid p \pm 1,$$

and let G be a group of order p^2q^2 . Show that G is abelian.

January 2017 Qualifier, Problem 2

Determine the number of Sylow p -subgroups in S_p .

August 2017 Qualifier, Problem 3

Suppose p and q are distinct primes. Show that if

$$p \nmid q - 1 \quad \text{and} \quad q \nmid p \pm 1,$$

then every group of order p^2q is abelian.

January 2018 Qualifier, Problem 8

Let p be a prime, let G be a finite group whose order is divisible by p , and let P be a normal p -subgroup of G . Show that P is contained in every Sylow p -subgroup of G .

August 2019 Qualifier, Problem 5

Let p, q, r be primes with $p < q < r$. Show that a group G of order pqr has a normal Sylow subgroup of order p , q , or r .

II. Closely related later-material problem

January 2018 Qualifier, Problem 9

Let p and q be distinct primes and let G be a group of order pq^r for some $r \geq 1$. If

$$q \nmid p-1 \quad \text{and} \quad p \nmid q^i-1 \quad \text{for every } 1 \leq i \leq r,$$

show that G is nilpotent. For what values of r must G be abelian?

January 2020 Qualifier, Problem 3

What is the order of a Sylow p -subgroup of S_{3p} for $p > 3$? Are the Sylow p -subgroups of S_{3p} abelian? Justify your answers.

August 2020 Qualifier, Problem 2

Consider $G = S_{28}$.

- (a) What is the order of a Sylow 7-subgroup of G ?
- (b) Is a Sylow 7-subgroup abelian? Justify your answer.

Source

University of New Mexico, Department of Mathematics and Statistics, [Past Qualifying Exams – Algebra](#). Dates and problem numbers are recorded above.